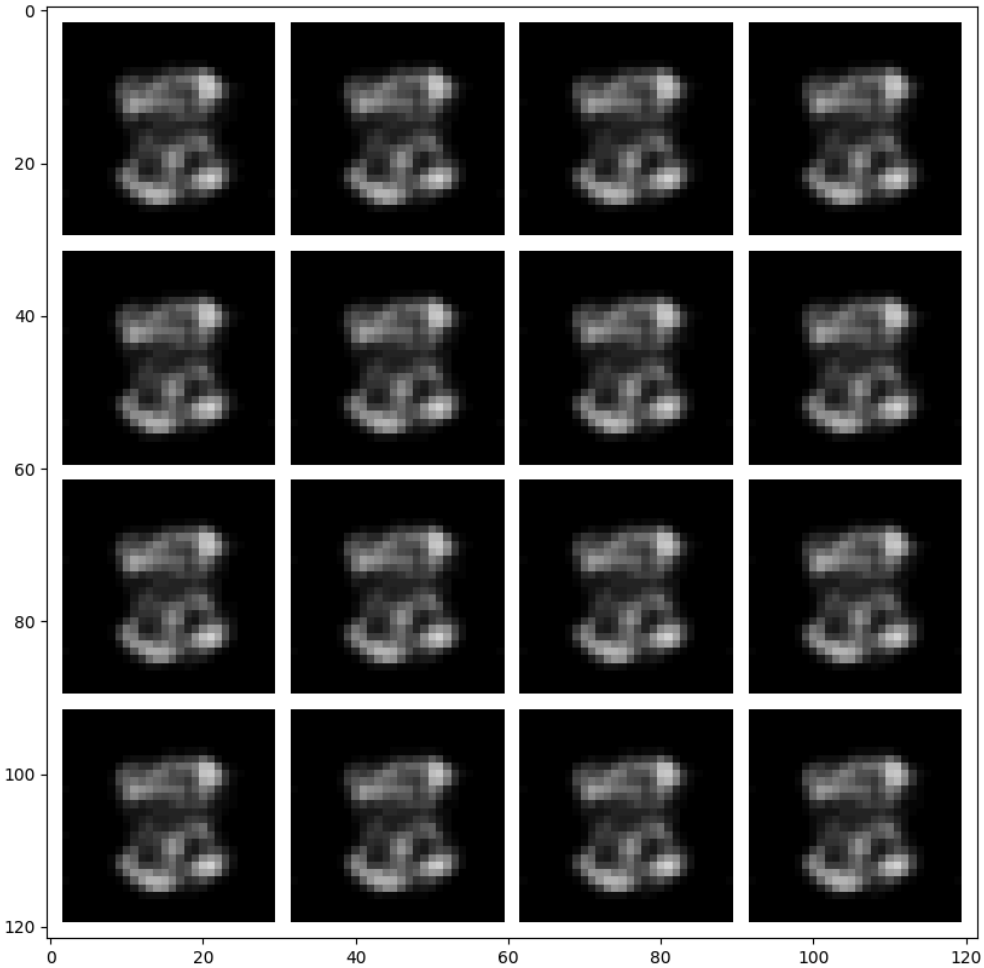
Autoencoder Samples



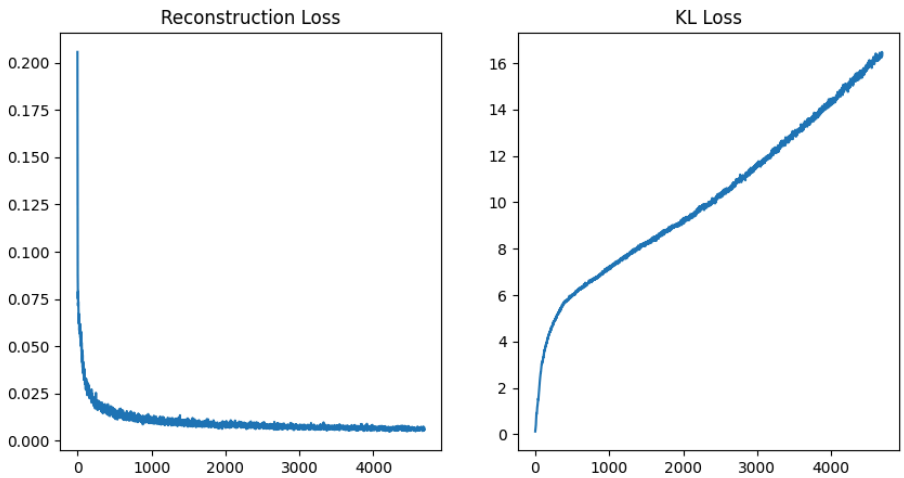
AE Sampling Inline Question Answer

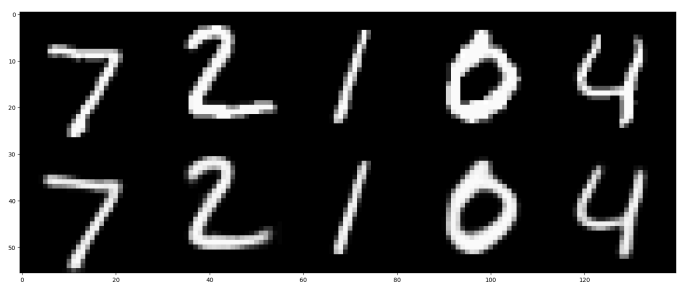
Q: Describe your observations, why do you think they occur?

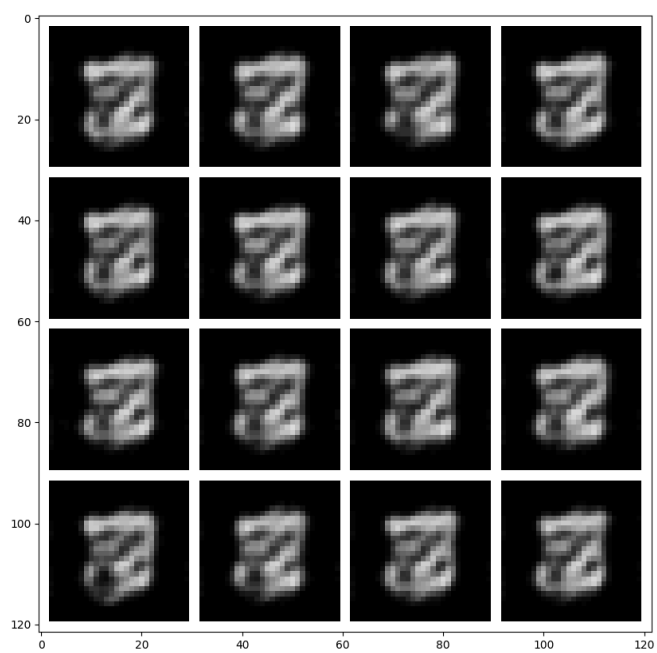
The resulting images are very irregularly shaped and unlike any of the input images. This is due to the fact that plain autoencoders have no constraints on where in the latent space they encode inputs. This usually results in sparsely spread out regions that can also be irregularly shaped. As a result, when we randomly sample, there is a high likelihood that it was sampled from a region that was never seen during the autoencoder's training. As a result, the autoencoder has not learned how to handle this sample, and thus produces an incoherent image.

VAE Training Curves and Reconstruction Samples

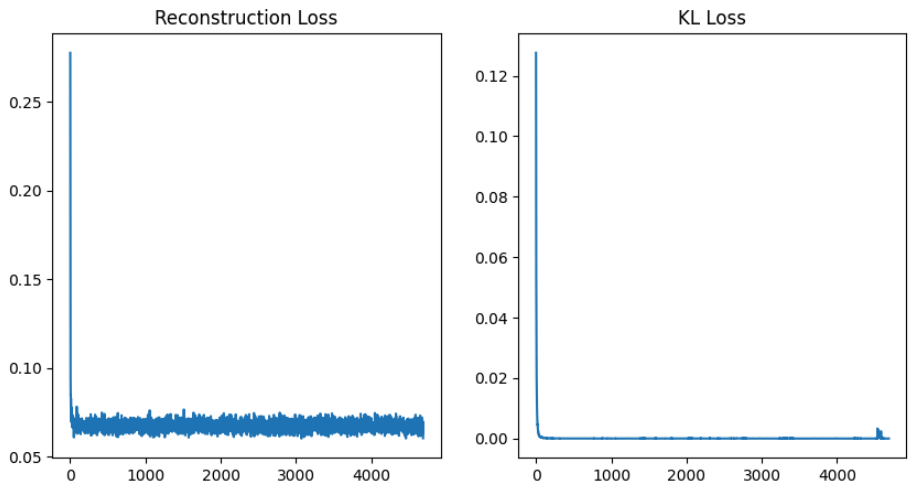
- **Beta = 0**

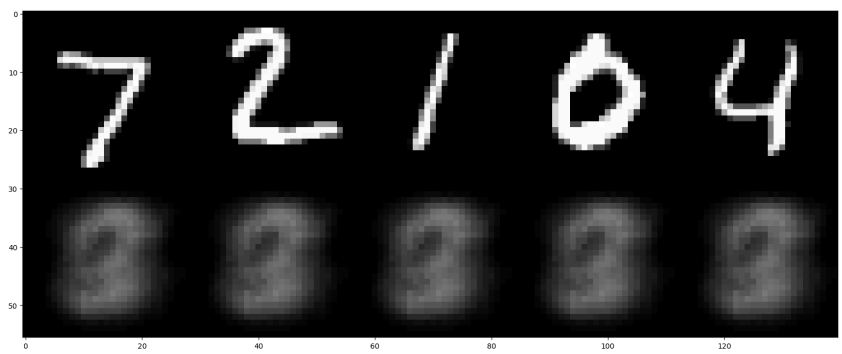


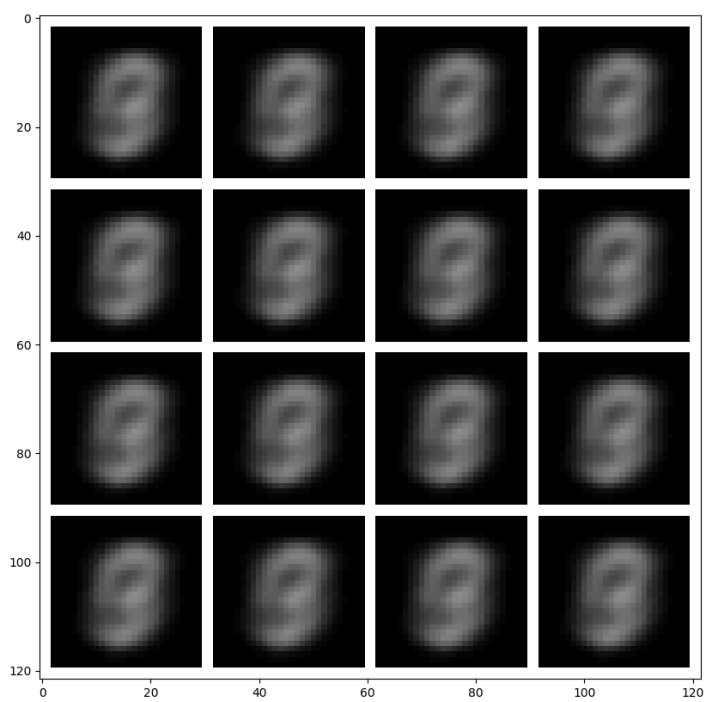




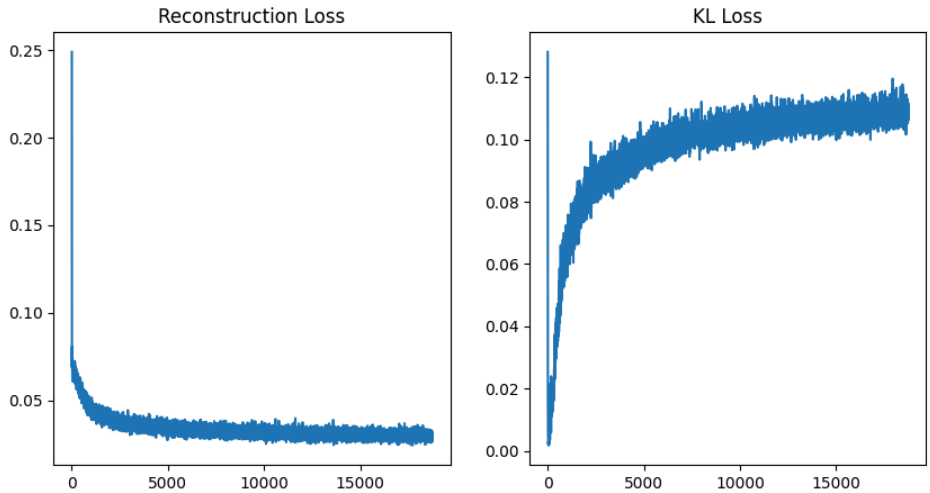
- **Beta = 10**

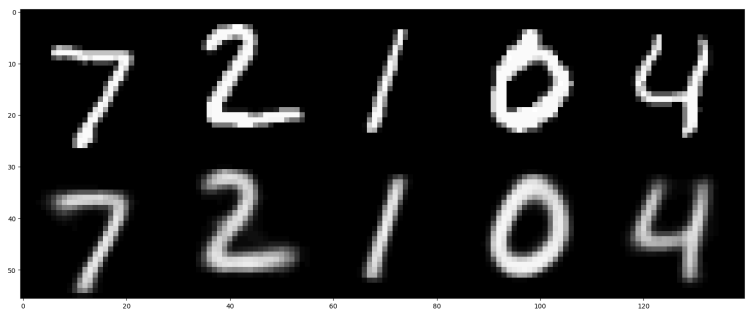

Reconstruction Loss
0.25
0.20
0.15
0.10
0.05
0
1000
2000
3000
4000
KL Loss
0.12
0.10
0.08
0.06
0.04
0.02
0.00
0
1000
2000
3000
4000

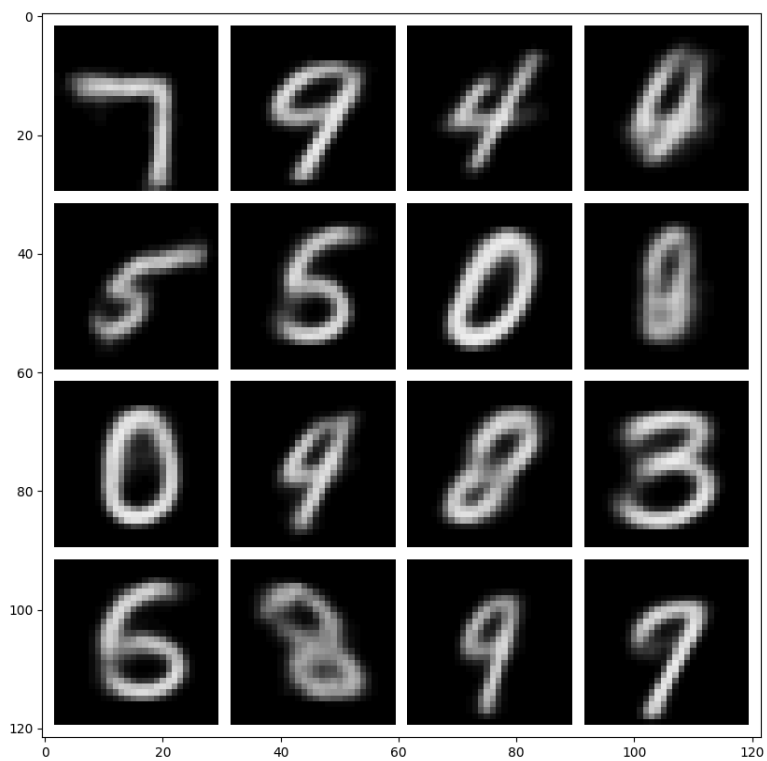




- **Tuned Beta (tuned value for Beta = 0.15)**







Answers to VAE Inline Questions

Q: What can you observe when setting β=0? Explain your observations!

While the reconstructions are good, the samples are again entirely incoherent. This is due to the fact that when the beta parameter is set to 0, the KL loss term disappears and the VAE essentially becomes a plain autoencoder. There is no constraint to align the latent space with the normal distribution.

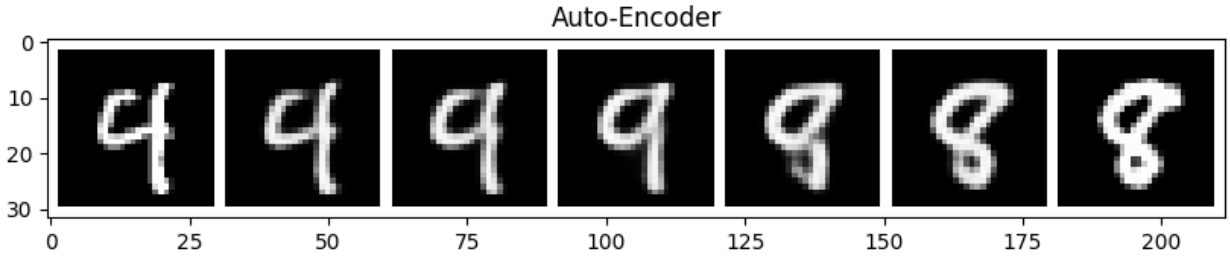
Q: What can you observe when setting β=10? Explain your observations!

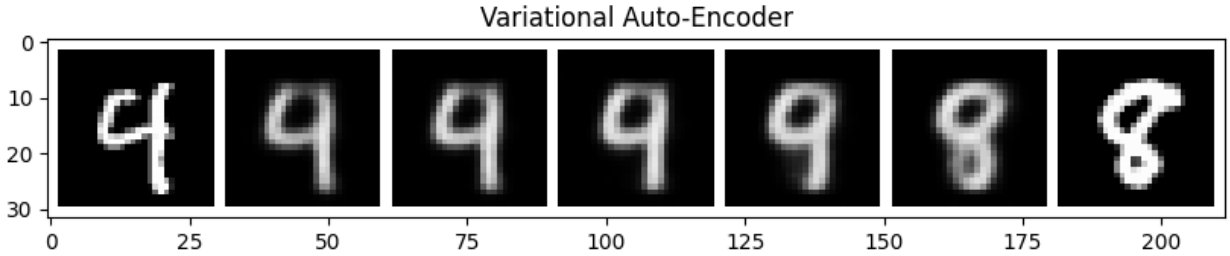
Both the reconstruction and sampled outputs have become blurry and unrecognizable. This is due to the fact that a beta value of 10 is too aggressively scaling the KL-loss term, and as a result every encoding is being forced heavily to be within the normal distribution. As a result, much of input-specific information is lost and encodings begin to all look similar, regardless of the original input.

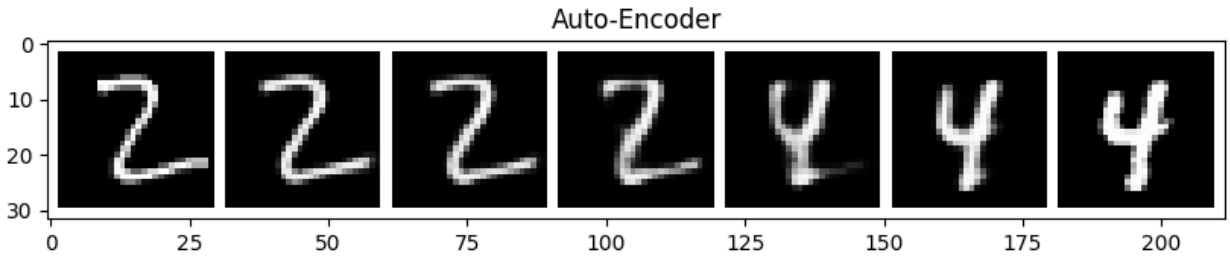
Q: Characterize what properties you would expect for reconstructions (1pt) and samples (2pt) of a well-tuned VAE!

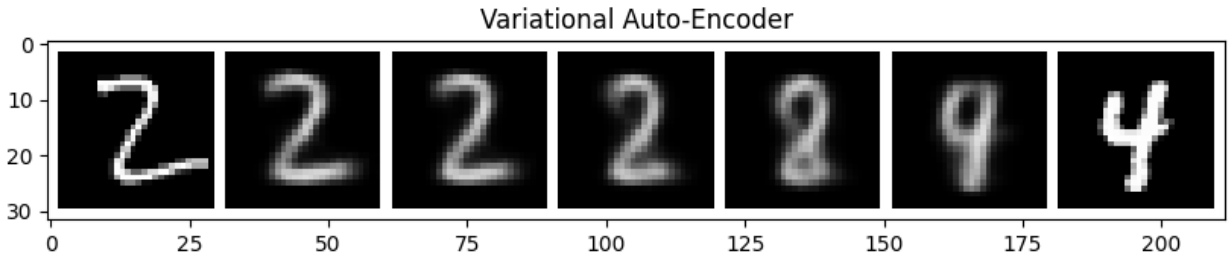
We would expect clear and accurate reconstructions, as well as clear, recognizable, and distinct samples from a well-tuned VAE.

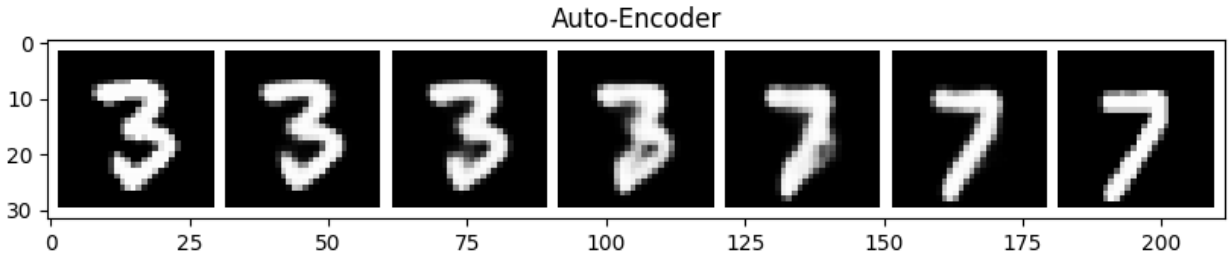
Three AE vs. VAE Representative Interpolation Comparisons

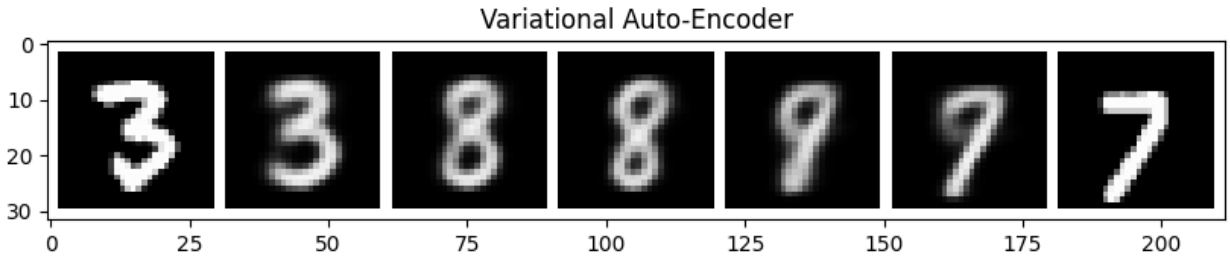












Interpolation Inline Question Answer

Q: How do AE and VAE embedding space interpolations differ?

Q: How do you expect these differences to affect the usefulness of the learned representation for downstream learning?

The embedding space interpolations differ for Auto-Encoder and Variational Auto-Encoders in the intermediate representations. For the Auto-Encoder, many of them are not “clean” representations of the data (handwritten digits), and the transitions

between start and end labels are abrupt. For the Variational Auto-Encoder, the transition is much smoother, and even the intermediate representations are actual “clean” representations of known handwritten digit labels. This is due to the fact that for Variational Auto-Encoders, the KL-loss term forces the latent space to be continuous and organized around the normal distribution. As a result, similar-looking digits are grouped closer together, and in a continuous fashion.

For the reasons described above, the representations of the Variational Auto-Encoder are far more useful, since they give way to better generalization. Because representations of similar inputs are grouped closer together, a classifier (or any other downstream learning) trained on this latent space will perform more reliably on unseen inputs.